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CURRENT POSITION

2024 - 2028 **PhD Candidate in Applied Mathematics**, fully funded by an IPAL–A*STAR fellowship.

Enrolled at: Université de Toulouse (École Doctorale EDMITT), Institut de Mathématiques de Toulouse (IMT), France.

Currently based at: Bioinformatics Institute (BII), A*STAR & IPAL (CNRS IRL 2955), Singapore — since April 2026.

Thesis: *Large-scale reconstruction methods for high-quality 3D photoacoustic imaging.*

Supervisors: Dr. Caroline Chaux, Dr. Paul Escande, and Dr. Hwee Kuan Lee.

- Developing fast and accurate reconstruction methods — combining inverse problems, optimization, and deep learning — to deploy a novel 3D photoacoustic tomography (PAT) scanner for biomedical imaging.
- Research carried out between IMT, Toulouse (2024 – early 2026) and BII, A*STAR / IPAL, Singapore (from April 2026).

EDUCATION

2024 - 2028 **PhD in Applied Mathematics** (in progress), Université de Toulouse, France — joint program with A*STAR, Singapore.

Inverse problems, computational imaging, and deep learning for 3D photoacoustic tomography.

2021 - 2024 **Engineering Degree (Master 2) in Applied Mathematics**, INSA Toulouse, France. Department of Mathematical & Modeling Engineering (GMM).

Focus on statistics, optimization, and machine learning; coursework in inverse problems, signal/image processing, and high-dimensional data analysis.

2019 - 2021 **Preparatory Classes for Grandes Écoles (CPGE)**, Hue University of Education, Vietnam.

Two-year intensive program in mathematics and physics, preparing for the competitive entrance examinations of French engineering schools.

2016 - 2019 **Baccalaureate in Mathematics**, Quoc Hoc - Hué High School for the Gifted, Vietnam.

Specialized mathematics track; graduated with honors.

PROFESSIONAL EXPERIENCE

May - June 2025 Visiting Research Stay — Photoacoustic Image Reconstruction

*Bioinformatics Institute (BII), IPAL, A*STAR, Singapore*

Supervisor: Dr. Hwee Kuan Lee

- Started a collaboration with the A*STAR team on deep learning approaches for 3D photoacoustic tomography (PAT) image reconstruction.
- Explored learning-based reconstruction for large-scale 3D PAT data, laying the groundwork for the Singapore phase of the PhD.

Mar - Aug 2024 Internship — Image Processing & Computer Vision

Cerema Occitanie, France

Supervisors: Dr. Guillaume Saint Pierre, Dr. Pascal Salmane

- Analyzed road-user behavior and interactions from aerial / smart-camera data using object detection and trajectory extraction.
- Built a computer-vision pipeline (Python, deep learning) to estimate trajectories for road-safety risk analysis.
- Co-authored a paper accepted at the IEEE ICPRS-24 conference.

June - Sept 2023 Internship — Image Processing & Inverse Problems

Institut de Mathématiques de Toulouse (IMT), France

Supervisor: Dr. Paul Escande

- Implemented optimization algorithms to solve inverse problems in photoacoustic tomography.
- Developed a blood-vessel image simulator to generate synthetic training data.
- Trained deep neural networks for image reconstruction using PyTorch.

TEACHING EXPERIENCE

2025 - 2026 Teaching Assistant — Optimisation (M1 Graduate Course)

Université Toulouse III - Paul Sabatier (24h TD, 12h TP)

Led tutorials and lab sessions on continuous and convex optimization: gradient and descent methods, constrained optimization and KKT conditions, with numerical implementation in Python.

2024 - 2025 Teaching Assistant — Fourier Analysis and Signal Theory (L3 Undergraduate Course)

Université Toulouse III - Paul Sabatier (30h TD)

Led tutorials on Fourier series and transforms, sampling and convolution, and their application to signal filtering and spectral analysis.

PUBLICATIONS & PRESENTATIONS

Publications

1. **T-T. Do**, C. Chaux, P. Escande, J. Gâteau, “Reconstruction d’images en tomographie photoacoustique avec régularisation combinée variation totale - Cauchy,” *Published at XXXe Colloque GRETSI 2025*.
2. J. M. Rico Velázquez, P. H. Salmane, G. Saint-Pierre, L. Khoudour, **T. T. Do**, E. Martin, “Evaluation of road users trajectories from aerial-view data for risk analysis,” *Published at 14th International Conference on Pattern Recognition Systems (ICPRS)*, 2024.

In Preparation

1. **T-T. Do**, et al., “Implementations of photoacoustic tomography models,” (*in preparation*).
2. **T-T. Do**, et al., “Electrical impulse response for 3D photoacoustic tomography models,” (*in preparation*).

Oral Presentations & Posters

- **Poster**, *Large-scale reconstruction methods for high-quality 3D photoacoustic imaging*, SinFra 2026, Singapore (Jun. 2026).
- **Poster**, *Journées modèles proxy et co-conception*, Institut Henri Poincaré, Paris, France (Sept. 2025).
- **Talk**, *XXXe Colloque GRETSI*, Strasbourg, France (Aug. 2025).

TECHNICAL SKILLS & LANGUAGES

<i>Programming</i>	Python, C/C++, CUDA, MATLAB, R.
<i>Scientific computing</i>	PyTorch, NumPy/SciPy, Scikit-Learn, Matplotlib.
<i>Tools</i>	Git, Docker, Linux, L ^A T _E X, Jupyter.
<i>Languages</i>	Vietnamese (native), English (professional), French (fluent).

OTHER INTERESTS

<i>Sports</i>	Football, racket sports, running.
<i>Music</i>	Guitar, pop music.
<i>Other</i>	Travelling, photography.